THE HARDY–STEIN IDENTITY FOR VERTICAL LIMITS AT THE ENDPOINT $p = 1$

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Abstract. Andersson proved a Hardy–Stein identity for the vertical limits f_χ of a Dirichlet series f in the Hardy space $\mathscr{H}^p$, for every χ in the p-Carlson set of f when $1 < p < \infty$, and for χ in a different set of full measure when $p = 1$. He asked whether the full Carlson set suffices also for $p = 1$. We show that it does. The only change to his argument is that uniform integrability of the derivatives, which follows from Riesz-mean approximation, replaces an L^2 bound in the comparison of $\partial_\sigma|f|$ with its Riesz means.

1. Introduction

For a Dirichlet series g and $\sigma > 0$ write

$$M_T(q)(\sigma) = \frac{1}{2T}\int_{-T}^{T} q(\sigma + it)\,\mathrm{d}t,$$

for functions q on the right half-plane, and let $R_N^k g$ be the Riesz means of g. Following Andersson [1, Section 2], for a Dirichlet series f and $1 \le p < \infty$, the *p-Carlson set* $\mathscr{C}_p(f)$ is the set of characters $\chi \in \mathbb{T}^\infty$ such that $f_\chi(s) = \sum_n a_n\chi(n)n^{-s}$ has an analytic continuation to $\operatorname{Re} s > 0$ with

$$\sup_{\sigma>0}\sup_{T\ge 1} M_T(|f_\chi|^p)(\sigma) < \infty.$$

If $f \in \mathscr{H}^p$, then $\mathscr{C}_p(f)$ has full measure [1, Theorem 2.1]. Andersson proved the following Hardy–Stein identity [1, Theorem 5.1]: if $f \in \mathscr{H}^p$, there is a set $E \subseteq \mathbb{T}^\infty$ of full measure such that for every $\chi \in E$ the map $\sigma \mapsto M_p^p(f_\chi, \sigma) = \lim_{T\to\infty} M_T(|f_\chi|^p)(\sigma)$ is continuously differentiable on $(0, \infty)$, and

$$\partial_\sigma M_p^p(f_\chi, \sigma) = -\lim_{T\to\infty} \frac{p^2}{2T}\int_{-T}^{T}\int_{\sigma}^{\infty} |f_\chi(\nu + it)|^{p-2}|f_\chi'(\nu + it)|^2\,\mathrm{d}\nu\,\mathrm{d}t, \tag{1}$$

with the limit uniform in $\sigma > \kappa$ for every $\kappa > 0$. For $p \ne 1$ one may take $E = \mathscr{C}_p(f)$. For $p = 1$ his proof uses instead the locally uniform L^2 hypothesis $\sup_{\sigma>\kappa,T\ge1} M_T(|f_\chi|^2)(\sigma) < \infty$ for every $\kappa > 0$, which holds on the set $E = \bigcap_n \mathscr{C}_2(f_{1/n})$, $f_{1/n}(s) = f(s+1/n)$, and he asked the following [1, Question 1].

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Question 1 (Andersson)**.** *Can one take $E = \mathscr{C}_1(f)$ in the Hardy–Stein identity also when $p = 1$?*

Theorem 1. *Let g be a somewhere convergent Dirichlet series with an analytic continuation to $\operatorname{Re} s > 0$ satisfying $\sup_{\sigma>0} \sup_{T\geq 1} M_T(|g|)(\sigma) < \infty$. Then $\sigma \mapsto M_1(g,\sigma) = \lim_T M_T(|g|)(\sigma)$ is continuously differentiable on $(0,\infty)$ and satisfies (1) with $p = 1$ and $f_\chi = g$, with the limit uniform in $\sigma > \kappa$ for every $\kappa > 0$.*

If $f \in \mathscr{H}^1$, then $|a_n| \leq \|f\|_{\mathscr{H}^1}$, so f_χ converges absolutely in $\operatorname{Re} s > 1$. Hence f_χ satisfies the hypothesis of Theorem 1 for every $\chi \in \mathscr{C}_1(f)$, and the answer to Question 1 is yes.

Corollary 1. *If $f \in \mathscr{H}^1$, then the Hardy–Stein identity (1) with $p = 1$ holds for every $\chi \in \mathscr{C}_1(f)$.*

We follow Andersson's proof [1, Section 5]. In his comparison of $\partial_\sigma |R_N^k g|$ with $\partial_\sigma |g|$ in [1, Lemma 5.9], an L^2 bound on g' controls a product of two small quantities through the Cauchy–Schwarz inequality. We show instead that the derivatives $(R_N^k g)'$ are uniformly integrable on vertical segments, uniformly in σ, and split the exceptional set in two. To keep the note self-contained we also treat the Dirichlet polynomials $R_N^k g$ directly (Lemma 3).

2. Uniform integrability

Throughout, g is as in Theorem 1, $k > 3$ is fixed, and C_0 denotes the supremum in its hypothesis. We use three results of Andersson for such g, all with $p = 1$:

(A1) By [1, Lemma 2.3], for every $\kappa > 0$,

$$\sup_{\sigma>\kappa} \sup_{T\geq 1} M_T(|g'|)(\sigma) \leq \frac{(1+\kappa)C_0}{\kappa}.$$

(A2) By [1, Lemma 2.4(c)], for every $\kappa > 0$,

$$\lim_{N\to\infty} \sup_{\sigma>\kappa} \sup_{T\geq 1} M_T(|R_N^k g - g|)(\sigma) = 0.$$

(A3) By [1, Lemma 4.10], if $g \not\equiv 0$, then for $0 < \alpha < \beta$,

$$\lim_{\delta\downarrow 0} \limsup_{T\to\infty} \sup_{\alpha<\sigma<\beta} \frac{|\{t \in [-T,T]; |g(\sigma+it)| \leq \delta\}|}{2T} = 0.$$

Lemma 1. *Let $\kappa > 0$. Then $\sup_{\sigma>\kappa, T\geq 1} M_T(|(R_N^k g)' - g'|)(\sigma) \to 0$ as $N \to \infty$. Moreover, for every $\varepsilon > 0$ there is $\eta > 0$ such that*

$$\frac{1}{2T}\int_E |h(\sigma+it)|\,\mathrm{d}t < \varepsilon$$

for every $\sigma > \kappa$, $T \geq 1$, every measurable $E \subseteq [-T,T]$ with $|E| \leq 2T\eta$, and every h in the family $\{g'\} \cup \{(R_N^k g)'; N \geq 2\}$.

Proof. Put $h_0(s) = g'(s + \kappa/2)$. By (A1) with $\kappa/2$ in place of κ, h_0 satisfies the hypothesis of Theorem 1, and it is a somewhere convergent Dirichlet series. Since $R_N^k h_0(s) = (R_N^k g)'(s + \kappa/2)$, (A2) applied to h_0 with $\kappa/2$ gives the first claim.

For the second, let $\varepsilon > 0$ and choose m such that $\sup_{\sigma>\kappa, T\geq 1} M_T(|(R_N^k g)' - g'|)(\sigma) < \varepsilon/3$ for every $N \geq m$. The Dirichlet polynomials $(R_N^k g)'$, $2 \leq N \leq m$, are bounded on $\operatorname{Re} s > \kappa$ by some $L > 0$; let $\eta = \varepsilon/(3L)$. For $N < m$ the integral over E is at most $L\eta = \varepsilon/3$. For $h = g'$ or $h = (R_N^k g)'$ with $N \geq m$, we have $|h| \leq |(R_m^k g)'| + |h - g'| + |(R_m^k g)' - g'|$, so the integral over E is less than $L\eta + 2\varepsilon/3 = \varepsilon$. □

3. Comparison of the derivatives

For an analytic function q put $D(q) = \partial_\sigma |q| = \operatorname{Re}(q'\overline{q})/|q|$ off the zeros of q, and $D(q) = 0$ at its zeros. Then $|D(q)| \leq |q'|$. For $a, b \neq 0$ and $A, B \in \mathbb{C}$, writing $u_a = a/|a|$, we have $|u_a - u_b| \leq 2|a - b|/|b|$, and hence

$$|\operatorname{Re}(A\overline{u_a}) - \operatorname{Re}(B\overline{u_b})| \leq |A - B| + \frac{2|B||a - b|}{|b|}. \tag{2}$$

Lemma 2. *Let $\kappa > 0$ and $g \not\equiv 0$. Then*

$$\lim_{N\to\infty} \limsup_{T\to\infty} \sup_{\sigma>\kappa} M_T(|D(R_N^k g) - D(g)|)(\sigma) = 0.$$

Proof. A half-plane far to the right. Let n_0 be the smallest index of a nonzero coefficient c of g, and write $g = n_0^{-s} F$ and $R_N^k g = n_0^{-s} F_N$. Choose $\beta > \kappa$ to the right of the abscissa of absolute convergence of g such that $|F - c| < |c|/4$ on $\operatorname{Re} s \geq \beta$. The functions $F_N - F$ and $F_N' - F'$ are Dirichlet series in the variables $(n/n_0)^{-s}$, $n \geq n_0$, whose coefficients are those of F and F' times $(1 - \log n/\log N)_+^k - 1$; by dominated convergence they tend to 0 uniformly on $\operatorname{Re} s \geq \beta$, where F' is bounded. Hence $|F_N| \geq |c|/2$ there for large N, and by (2) $|D(F_N) - D(F)| \leq |F_N' - F'| + 2|F'||F_N - F|/|F| \to 0$ uniformly. Since $D(n_0^{-s} F) = n_0^{-\sigma}(-\log n_0 |F| + D(F))$, we get $D(R_N^k g) \to D(g)$ uniformly on $\operatorname{Re} s \geq \beta$.

The strip $\kappa < \sigma < \beta$. Let $\varepsilon > 0$, and let η be as in Lemma 1. By (A3) choose $\delta > 0$ with

$$\limsup_{T\to\infty} \sup_{\kappa<\sigma<\beta} \frac{|\{t; |g(\sigma + it)| < \delta\}|}{2T} < \frac{\eta}{3},$$

and then choose $0 < \rho < \delta/2$ with $4\rho(1 + \kappa)C_0/(\kappa\delta) < \varepsilon$. Fix σ and T, write g, g_N for $g(\sigma + it)$, $R_N^k g(\sigma + it)$, and split $[-T, T]$ into

$$E_1 = \{|g| < \delta\}, \quad E_2 = \{|g_N - g| > \rho\} \setminus E_1, \quad E_3 = [-T, T] \setminus (E_1 \cup E_2).$$

On E_3 we have $|g| \geq \delta$ and $|g_N - g| \leq \rho < \delta/2$, so $g_N \neq 0$, and (2) gives $|D(g_N) - D(g)| \leq |g_N' - g'| + (2\rho/\delta)|g'|$. On $E_1 \cup E_2$ we use $|D(g_N) - D(g)| \leq |g_N'| + |g'|$. By Markov's inequality and (A2), $|E_2|/(2T) \leq \rho^{-1} M_T(|g_N - g|) <$

$\eta/3$ for all $\sigma > \kappa$, $T \geq 1$ once N is large. So for large N and large T, the set $E_1 \cup E_2$ has measure at most $2T\eta$, and Lemma 1 and (A1) give

$$\sup_{\kappa<\sigma<\beta} M_T(|D(g_N) - D(g)|) \leq \sup_{\sigma>\kappa} M_T(|g_N' - g'|) + \frac{2\rho}{\delta}\frac{(1+\kappa)C_0}{\kappa} + 2\varepsilon.$$

By Lemma 1 the first term tends to 0 as $N \to \infty$, and the second is less than ε. Combined with the uniform convergence on $\operatorname{Re} s \geq \beta$, this proves the lemma. □

4. Dirichlet polynomials

At $p = 1$ the function $D(P)$ is not a continuous function of (P, P') at the zeros of P, so its vertical means need a short argument also for Dirichlet polynomials.

Lemma 3. *Let $P \not\equiv 0$ be a Dirichlet polynomial and $\kappa > 0$. Then the limits $u_P(\sigma) = \lim_T M_T(|P|)(\sigma)$ and $v_P(\sigma) = \lim_T M_T(D(P))(\sigma)$ exist uniformly in $\sigma > \kappa$, u_P is continuously differentiable on (κ, ∞), and $u_P' = v_P$.*

Proof. On $\operatorname{Re} s > \kappa$ the functions P and P' are bounded and almost periodic, so u_P exists uniformly in $\sigma > \kappa$ [1, Section 2]. For fixed σ, $P(\sigma + it) = 0$ for only finitely many $t \in [-T, T]$, and the difference quotients of $\sigma \mapsto |P(\sigma+it)|$ are bounded by $\sup|P'|$. By dominated convergence $M_T(|P|)$ is differentiable with derivative $M_T(D(P))$, and $M_T(D(P))$ is continuous in σ.

Far to the right, write $P = n_0^{-s}Q$ as in the proof of Lemma 2, with $|Q| \geq |c|/2$ on $\operatorname{Re} s \geq \beta$. There $D(P) = n_0^{-\sigma}(-\log n_0|Q| + \operatorname{Re}(Q'\overline{Q})/|Q|)$ is a uniformly continuous function of the almost periodic pair (Q, Q'), so its means converge uniformly in $\sigma \geq \beta$. On the strip $\kappa < \sigma < \beta$, let $D_\varepsilon(P) = \operatorname{Re}(P'\overline{P})/\max(|P|, \varepsilon)$. This is a uniformly continuous function of (P, P'), so its means converge uniformly in σ, and $|D(P) - D_\varepsilon(P)| \leq 2\sup|P'|\,\mathbf{1}_{\{|P|<\varepsilon\}}$. By (A3) for P, the density of $\{|P| < \varepsilon\}$ is small, uniformly on the strip, for large T and small ε. Hence $M_T(D(P))$ is uniformly Cauchy as $T \to \infty$, and v_P exists uniformly in $\sigma > \kappa$; it is continuous as a uniform limit of continuous functions. Finally $M_T(|P|)(\sigma) - M_T(|P|)(\sigma_0) = \int_{\sigma_0}^{\sigma} M_T(D(P))$, and letting $T \to \infty$ gives $u_P' = v_P$. □

5. Proof of Theorem 1

We may assume that $g \not\equiv 0$. Fix $\kappa > 0$, let $g_N = R_N^k g$, and drop the finitely many N with $g_N \equiv 0$. By Lemma 3, $u_N = u_{g_N}$ and $v_N = v_{g_N}$ exist uniformly in $\sigma > \kappa$, and $u_N' = v_N$.

Let $A_N = \limsup_T \sup_{\sigma>\kappa} M_T(|D(g_N) - D(g)|)(\sigma)$, which tends to 0 by Lemma 2. Then $\sup_{\sigma>\kappa}|v_N - v_{N'}| \leq A_N + A_{N'}$, so v_N converges uniformly to some v, and

$$|M_T(D(g)) - v| \leq M_T(|D(g) - D(g_N)|) + |M_T(D(g_N)) - v_N| + |v_N - v|$$

gives $\limsup_T \sup_{\sigma>\kappa}|M_T(D(g)) - v| \leq A_N + \sup_{\sigma>\kappa}|v_N - v| \to 0$. So $\lim_T M_T(D(g)) = v$ uniformly in $\sigma > \kappa$. By (A2), $M_T(|g|)$ and $M_T(|g_N|)$

differ by at most $\sup M_T(|g_N - g|)$, which tends to 0 uniformly in $T \geq 1$ and $\sigma > \kappa$; hence $M_1(g, \sigma) = \lim_T M_T(|g|)$ exists, uniformly in σ, and $u_N \to M_1(g, \cdot)$ uniformly. Since also $u_N' = v_N \to v$ uniformly, $M_1(g, \cdot)$ is continuously differentiable with derivative v [2, Theorem 7.17].

Finally, [1, Lemma 5.4] with $p = 1$, whose hypothesis is that of Theorem 1, shows that $M_T(D(g))$ and $-(2T)^{-1} \int_{-T}^{T} \int_{\sigma}^{\infty} \Delta|g|$ differ by a quantity that tends to 0 uniformly in $\sigma > \kappa$. Finally, $\Delta|g| = |g|^{-1}|g'|^2$ off the zeros of g, as in [1, Theorem 5.5]; near a zero of order m this function is of order $|s - s_0|^{m-2}$, hence locally integrable, and the Riesz measure of $|g|$ has no atom there, since $|\nabla|g|| = |g'|$ is bounded and the flux through small circles tends to 0. This gives (1) with $p = 1$. □

6. Concluding remark

The proof of [1, Lemma 5.9] controls the small values of f through the strip estimate (A3). When $f(+\infty) = 0$, the far right half-plane needs a separate argument, which the first part of the proof of Lemma 2 supplies. Since $D(P)$ is discontinuous at the zeros of P, we have also given the case of Dirichlet polynomials in full in Lemma 3.

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References

[1] Viktor Andersson, *The Littlewood–Paley formula and mean counting function for vertical limits of Dirichlet series* (2026), available at `arXiv:2606.20293`. ↑1, 2, 4, 5

[2] Walter Rudin, *Principles of mathematical analysis*, 3rd ed., McGraw-Hill, 1976. ↑5

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